Computer Science & IT



C Programming

Control Flow Statement

Lecture No. 06



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Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

Mathematical Aspect of for loop

Slide

Topics to be Covered

Control flow



- Topic — Nested for loop
- Topic — Some program
- Topic — Break , continue
- Topic — While and do while
- Topic

Slide



for loop

129/64/509.

Iteration

$2^0 = 1 <= n$ (1)

$2 <= n$ (2)

$2^2 <= n$ (3)

⋮ ⋮

(K)

$2^K <= n$

```
int i;
for (i=1; i<=n; i=i*2) {
    printf ("I am a good student");
}
```

variable

2^0 - extra count

No. of time loop execute $\lfloor \log_2 n \rfloor + 1$

Slide



for loop

```
for(i=2; i<=n; i=i*2)
```

$\lfloor \log_2 n \rfloor$

Iteration

1 $2^1 <= n$ ✓

2 $2^2 <= n$ ✓

3 $2^3 <= n$ ✓

.

.

.

K $2^K <= n$ ✓

↑ No. of Iteration

Slide



for loop

n = 10

```
for (i=1; i<=10; i=(i*2)) {
printf ("I am a good student");
}
```

Iteration

$1 <= 10$ ✓ ①

$2 <= 10$ ✓ ②

$2^2 <= 10$ ✓ ③

$2^3 <= 10$ ✓ ④

$2^4 <= 10$ ✗

$$\lfloor \log_2 n \rfloor + 1$$

$$\lfloor \log_2 10 \rfloor + 1 = 3 + 1 = 4$$

$3 < \quad < 4$

Slide



for loop

```
# include<stdio.h>
int main() {
  int i, j=1; Sum=0;
  for (i=1; i<=10; i++) {
      Sum= Sum+(j*j);
      j=j+1;
  }
  printf("%d", Sum);
  return 0;
}
```

1 — $0+1^2$

2 — $0+1^2+2^2$

3 — $0+1^2+2^2+3^2$

4

5

6

7

8

9

10 : $0+1^2+2^2+3^2+\ldots+10^2$

Slide



for loop

Sum = $1^2 + 2^2 + 3^2 + \cdots + 10^2$

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\frac{\overset{5}{\cancel{10}} \times 11 \times \overset{7}{\cancel{21}}}{\cancel{6}_{\cancel{2}}} = 385$$

Slide



for loop

Nested loop (Loop inside the loop)

```
int i,j;
for(i=1; i<=3; i++) {
    for(j=1; j<=3; j++) {
        printf("I am a good student");
    }
}
```

9 times

Inside block of first loop

i=1 → j=1, j=2, j=3 — printf — printf — printf

i=2 → j=1, j=2, j=3 — printf — printf — printf

i=3 → j=1, j=2, j=3 — printf — printf — printf

Slide



for loop

```
int i, j;
for (i=1; i<=10; i++) {
    for(j=1; j<=2, j++) {
        printf("I am a good student");
    }
}
```

(A) 10

(B) 2

(C) 20

(D) 100

i=1 j=1
j=2

i=2 j=1
j=2

⋮

i=10, j=1
j=2

10 * 2 = 20



for loop

NAT

ub - lb + 1 = 10 - (-5) + 1 = 16

```
int i, j
for (i = -5; i <= 10; i++) {
        for(j = 6; j <= 9; j++) {
            printf ("I am tired");
        }
}
```

9 - 6 + 1 = 3 + 1 = 4 times

16 * 4 = 64

Slide

for loop

```
int i,j;
for(i=10; i<=20; i++){
    for(j=0; j++; j=j+1){
        printf("I am brilliant");
    }
}
```

20 - 10 + 1 = 11

No print

0

0

✓

post increment

Second loop

false

(A) Infinite

(B) 1

(C) 20

(D) ~~0~~

Slide



for loop

```
int i, j;
for (i=1; i<=4; i++) {
    for(j=1; j<=4; j++) {
        printf("* \t");
    }
    printf("\n");
}
return 0
}
```

\t - tab keyboard

outer loop:

i = 1 j = 1 * * * *
j = 2
j = 3
j = 4

i = 2 j = 1 * * * *
j = 2 j = 3 j = 4

i = 3 * * * *
j = 1 j = 2 j = 3 j = 4

i = 4 * * * *
j = 1 j = 2 j = 3 j*

Slide



for loop

```
int i,j;
for (i=1; i<=n; i++) {
    for(j=1; j<=n; j++) {
        printf("*\t");
    }
    printf("\n");
}
```

total No. of time "*" printed is n^2

Slide

for loop

Imp

```
int i,j;
for(i=1; i<=4; i++){
    for(j=1; j<=i; j++){
        printf("* \t");
    }
    printf("\n");
}
```

No. of times * printed is?

40%

(A) 16

(B) 10

(C) 11

(D) 4

```
*
* *
* * *
* * * *
```

lide



for Loop

```
for(j=1; j<=i; j++)
```

j<=1

i=1 j=1 to 1 *

i=2 j=1 to 2 * *

i=3 j=1 to 3 * * *

i=4 j=1 to 4 * * * *

Slide



for loop

for any value of n.

Maths

$1+2+3+\cdots+n$

$\frac{n(n+1)}{2}$

```
for (i=1; i<=n; i++) {
    for(j=1; j<=i; j++) {
        printf("* \t");
    }
    printf("\n");
}
```

No. of * printed.

```
*
* *
* * *
* * * *
  :
* * * * - - *
```

Slide



for loop

```
# include<stdio.h>
int main(){
    int x=1, i;
    for(i=1; i<=5; i++){
        x=x*i;
    }
    printf("%d", x);
    return 0;
}
```

5 time

program output is?

i=1 x=1*1=1

i=2 x=1*2=2

i=3 x=2*3=6

i=4 x=6*4=24

i=5 x=24*5=120

output= 120

5! factorial Computation + Scanf + function

Slide



While loop

```
while ( condition) {
    stmt;
}
```

if condition evaluates to true then body of loop will execute

Non Zero - True ✓

Zero - false.

Slide



While loop

No. of time will execute

for loop.

you want to execute the body until condition is true then we prefer while loop

Condition dependency while

Slide



While loop

equivalent

```
for (expr1; expr2; expr3){ ⟹   expr1
                               while(expr2) {
    stmt;                          stmt;
    :                              :
}                                  expr3;
                               }
```

Slide



While loop

```
int i=4
while (i<=10) {
    printf("%d", i*i);
    i= i+1;
}
```

4<=10 — 16

5<=10 — 25

6<=10 — 36

7<=10, 49

8<=10. 64

9<=10 . 81

10<=10 100

11<=10 ✗

Slide



While loop

```
int i = 1
while (i != 10) {
    printf ("I am a good student");
    i = i + 1;
}
```

1 != 10 ✓
2 != 10 ✓
3 != 10
4 != 10
:
9 != 10 ✓
10 != 10 ✗

9 times

No. of times printf executed.

10 != 10
false

Slide



While loop

2014

```
# include<stdio.h>
int main() {
    int count=0, num=435;
    while(num) {
        count++;
        num= num >>1;
    }
    printf("%d", count);
    return 0;
}
```

(A) 9
(B) 10
(C) 8
(D) 0

True

0110110011

count=1, 0 0110110011

count=2, 0 0011011001

count=3 0 0001101100

Count 4 0 0000110110

Count 5, 0 0000011011

Count=6 0 0000001101

Count=7 0 0000000110

Count=8 0 0000000011

Count=9 0 0000000001

Slide

While loop

bit wise Complement



2	435	
2	217	1
2	108	1
2	54	0
2	27	0
2	13	1
2	6	1
2	3	0
2	1	1
	0	

0 1 1 0 1 1 0 0 1 1

↑ MSB Sign

435

count=1 $435/2 = 217$

count=2 108

count=3 54

count=4 27

Count=5 13

Count=6 6

Count=7 3

Count=8 1

count=9 0

Slide



break

break statement is used in switch and loop

break statement transfer control of loop or swith to out side of block in which its written.

Slide



break

No. of times printf statement executed is 41

```
int i, j
for (i=1; i<=3; i++) {
    for(j=1; j<=20; j++) {
        printf("I am a good student");
        if (i==2) break;
    }
}
```

i=1.

Control out of inner for loop

Slide



break

i=1. j=1,2,3... 20. 20 times

i=2 j=1. break; 1 time

i=3. j=1,2,3... 20 20 time

41 times

Slide



Continue

Continue dont break the loop

If continue statement encounter then rest statement in the loop is skipped.

retest the condition before reiterating.



Question: Continue

#Q.

```
#include<stdio.h>
void main () {
for (int i = 1; i <= 10; i++) {
if (i > 5) {
        break;
}
        printf("%d\n", i);
}
```

A 4
b.5
C.7
d. 11

Slide



Question: Continue

#Q.

```
#include<stdio.h>
void main () {
for (int i = 1; i <= 10; i++) {
if (i == 5||i==6) {
        continue;
}
        printf("%d\n", i);
}
}
```

A 4
b.5
C.8
d. 11

No. of time printf executed is 4

1 ✓
2 ✓
3 ✓
4 ✓
i=5 } Rest of part of loop will
i=6 } Not execute
7 ✓
8 ✓
9 ✓
10 ✓



Question: Continue

#Q.

```
#include<stdio.h>
void main () {
for (int i = 1; i <= 10; i++) {
if (i > 5) {
        continue;
}
        printf("%d\n", i);
}
```

A 4
b.5
C.10
d. 11

Slide



Question

#Q. Consider the following program

```
#include <stdio.h>
int main() {
        int i=2+4%6+9/10;
        while (i<10){
            printf("I am good student");
            i++;
        }
        return 0 ;
        }
```

The number of times printf statement executed is

A. 4
b. 5
C. 10
d. 11

$i = 2+4\%6+9/10$

$= 2+4+9/10$

$= 2+4+0 = 6$

$6<10$

$6<10$ ✓

$7<10$ ✓

$8<10$ ✓

$9<10$ ✓

$10<10$ ✗

Slide



Question

#Q.
```
#include <stdio.h>
int main() {
        int i=10;
        for (; i<=100; i++);{
                printf("\nI am a good Student");
        }
        printf("%d",i);
        return 0 ;
        }
```

The value of I is x and number of print statement is y
then value of x+y is



Question

#Q.

```
#include<stdio.h>

int main()

{

        int i=3;
        for(;i++<=9;++i);
        printf("%d",i);
        return 0;

}
```

The output is _____.

Slide

```
#include <stdio.h>
int main()
{
int a,i;
a= 12.5/5+31.2/2*5-5;
for(i=a; i<90;i=i+3)
    {
      printf("%c\t", i+32);
    }
    return 0;
}
```

The output is-

(a) 107 110 113 116 119

(b) K L M N O

(c) k n q t w

(d) Garbage values

$a = 2.5 + 15.6 * 5 - 5$

$a = 2.5 + 78.0 - 5$

$a = 80.5 - 5$

$a = 75.5$

75

75 ; i<90 ; i=i+3

75
32
107

97 - a
98 - b
99 - c
100 - d
101 - e
102 - f
103 - g
104 - h
105 - i
106 - j
107 - k

75 + 32 = 107

78 = 110

81 + 32 = 113

84 = 116

87 = 119



Question

#Q.

Consider the following program:

```
#include <stdio.h>
int main()
{
    int i, j, n;
    for(i=1;i<=n;i++)
    for(j=1;j<=n;j++)
    printf("GATE Wallah");
    return 0;
}
```

How many times will the printf() statement be executed when n=32?

(a) 128 (b) 1024
(c) 512 (d) 256

```
for(i=1; i<=32; i++){
    for(j=1; j<=32; j++):
        printf("Gate wallah"
```

n^2

$(2^5)^2 = 2^{10} = 1024$

Slide



Question

#Q. The number of times loop executed is

```
#include<stdio.h>
int main(){
int i,count=0;
for (i=1; i<=100; i=i*2 ){
     printf("\nI am a good Student");

}
return 0 ;
}
```

A. 6
B. 7
C. 8
D. 10

The number of times value will be printed?

Slide



2 mins Summary

Topic — for loop star panting

Topic — break

Topic — Continue

Topic — While loop.

Topic

Slide

THANK - YOU